

A Transient Alkaline Shift Reversibly Increases Association of the Chaperone ChpA with Heat-Damaged Lysozyme

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ABSTRACT

The fictional bacterial chaperone ChpA was examined for pH-dependent binding to heat-damaged lysozyme. Cell extracts prepared at pH 7.2, 8.0, or 8.5 were applied to lysozyme-affinity resin after a 10-min temperature challenge at 42 C. Binding at pH 8.5 was 3.2-fold higher than at pH 7.2 and returned to baseline after the extract was restored to pH 7.2 for 20 min. Total ChpA abundance did not change during the experiment. Phosphate incorporation and covalent modification were not assessed. The results demonstrate a reversible pH-dependent change in binding capacity but do not identify its molecular mechanism.

Keywords: *chaperone, alkaline shift, lysozyme, reversible binding, affinity capture*

INTRODUCTION

Molecular chaperones often respond to environmental conditions through changes in nucleotide state, oligomerization, or client availability. A reversible change in binding can therefore occur without new protein synthesis.

ChpA is a synthetic 68-kDa chaperone used here to test whether a short alkaline shift changes interaction with a damaged protein substrate. Heat-damaged lysozyme was selected as a standardized affinity ligand. The study was intentionally limited to functional binding and did not test phosphorylation, acetylation, or other covalent modifications.

MATERIALS AND METHODS

Extract preparation

A fictional Gram-negative strain expressing native ChpA was grown at 30 C. Cultures were shifted to 42 C for 10 min, rapidly chilled, and lysed in 25 mM HEPES adjusted to pH 7.2, 8.0, or 8.5. Sodium chloride was fixed at 150 mM in all conditions.

Lysozyme-affinity assay

Lysozyme was heated at 70 C for 15 min and coupled to activated agarose. Equal amounts of soluble cell protein were applied to 1-mL columns. Bound ChpA was eluted with 2 mM ATP and 5 mM MgCl2 and quantified by immunoblotting. Native lysozyme resin and uncoupled agarose were used as controls.

Reversal experiment

Extract adjusted to pH 8.5 was held for 10 min and then returned to pH 7.2. Aliquots were sampled immediately and after 5, 10, and 20 min. Total ChpA was measured in parallel to distinguish abundance from binding capacity.

Group	Challenge	Binding pH	Recovery step
Baseline	42 C, 10 min	7.2	None
Intermediate	42 C, 10 min	8.0	None
Alkaline	42 C, 10 min	8.5	None
Reversal	42 C, 10 min	8.5 -> 7.2	0-20 min

Table 1. Experimental groups.

RESULTS

Alkaline pH increased ChpA capture

Relative to the pH 7.2 baseline, ChpA recovered from damaged-lysozyme resin was 1.7-fold higher at pH 8.0 and 3.2-fold higher at pH 8.5. Binding to native lysozyme was low at all pH values, and uncoupled agarose retained less than 4% of the signal measured on damaged-lysozyme resin.

The effect was reversible

After the pH 8.5 extract was returned to pH 7.2, elevated binding declined over time. The relative signal fell from 3.2 at time zero to 2.2 at 5 min, 1.5 at 10 min, and 1.1 at 20 min. Total ChpA abundance varied by less than 6% across these samples.

Scope of the evidence

No radioactive phosphate labeling, phosphatase treatment, mass spectrometry, or mutation analysis was performed. The experiment therefore supports a reversible functional state but does not establish a covalent modification.

Condition	Relative ChpA binding	Total ChpA
pH 7.2	1.0	1.00
pH 8.0	1.7	0.98
pH 8.5	3.2	1.04
pH 8.5 -> 7.2, 20 min	1.1	1.01

Table 2. Binding and total abundance normalized to pH 7.2.

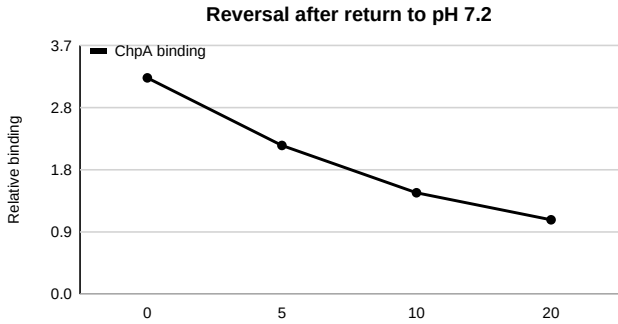


Figure 1. The alkaline-induced binding state decayed over 20 min after neutralization.

DISCUSSION

A brief alkaline shift increased ChpA association with damaged lysozyme without increasing total ChpA abundance. The rapid reversal after neutralization indicates that the altered state is not permanent.

Several mechanisms could produce this response, including nucleotide exchange, altered client exposure, or reversible modification. None was directly measured. In particular, the present data should not be summarized as evidence that ChpA was phosphorylated.

LIMITATIONS

The affinity ligand was an artificial damaged substrate, and the study did not identify the biochemical basis of the pH response. Only one heat challenge and one ionic-strength condition were tested.

CONCLUSION

ChpA binding to heat-damaged lysozyme increased 3.2-fold at pH 8.5 and returned nearly to baseline within 20 min after restoration to pH 7.2, without a change in total ChpA abundance.

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